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Lecture 12: Data Forensics and Evidence Security

i) Data Forensics

Data forensics is a specialized field within cybersecurity that focuses on the **investigation of digital crimes**. It involves the systematic collection, preservation, analysis, and presentation of digital evidence in a way that ensures it is admissible in a court of law.

This discipline is critical in uncovering facts related to incidents such as **cyberattacks, data breaches, intellectual property theft, and insider threats**. Forensic experts work to piece together digital trails left behind during such incidents to understand what occurred, how it happened, and who may have been responsible.

To achieve this, data forensics relies on several **technical procedures and tools**. One of the foundational techniques is **disk imaging**, which entails creating an exact replica of a storage device, such as a hard drive. This allows investigators to work on a duplicate without risking damage or alteration to the original evidence. For example, in a criminal investigation, a suspect's hard drive may be imaged so that its contents can be examined without compromising data integrity.

Another important process is **file recovery**, which focuses on retrieving deleted, hidden, or corrupted files that may contain crucial evidence. Tools like **Recuva** or **FTK Imager** are widely used in this context, as they can often recover information even from reformatted or damaged drives. Additionally, **log analysis** is used to examine system and network logs. This helps forensic analysts identify unauthorized access or track how a breach was carried out. For instance, examining login logs may reveal a pattern of brute-force attacks or suspicious access attempts.

ii) Evidence Security

Maintaining the **security and integrity of evidence** throughout the forensic investigation process is paramount. One of the most critical components in this regard is the **chain of custody**. This refers to the documented process that outlines who handled the evidence, when, where, why, and how. Each piece of evidence should have a **unique identifier**, and every transfer or access must be recorded with precise details such as the date, time, purpose, and personnel involved. This ensures that the **credibility and authenticity of the evidence** are preserved, which is especially important if the evidence is to be presented in court.

To prevent **evidence tampering or loss**, strict control measures must be implemented. This includes using **tamper-evident packaging** for physical storage devices, such as hard drives or USBs, to show any signs of unauthorized access. For digital evidence, **encryption and role-based access controls** are essential to prevent data breaches or manipulation. Furthermore, evidence storage facilities should be physically secured with **surveillance systems, access logs, and restricted entry** to authorized personnel only. These precautions collectively ensure that evidence remains uncompromised throughout the investigative lifecycle.

iii) Preservation Best Practices

Preserving digital evidence correctly is essential for both **legal validity and long-term usability**. One of the primary practices is ensuring **secure storage**. This involves keeping digital evidence in a **climate-controlled and physically secured environment**, which protects it from environmental damage, tampering, or unauthorized access. For example, sensitive storage devices should be placed in **fireproof safes** and locked in restricted-access rooms or cabinets.

Another best practice is to maintain **regular backups** of all digital evidence. Creating multiple copies of the data ensures that if one is accidentally damaged or lost, others are available for continued investigation. These backups should ideally be stored in separate, secure locations, such as a combination of on-site servers and cloud-based storage, to provide redundancy and additional security.

Equally important is the comprehensive documentation of all evidence handling activities. From the initial collection of the evidence to each phase of analysis, transfer, and storage, every action must be recorded in detail. These records should include the date and time of each step, the identities of personnel involved, the purpose of any access or modifications, and the outcomes of each phase. Using standardized forms or digital case management software helps streamline this process and ensures consistency and accountability across the entire investigation.